

Markscheme

Mathematics: analysis and approaches
Practice question bank

157. (a)

Given that $x = 1 - i$ is a root of $x^4 - 2x^3 - x^2 + 6x - 6 = 0$.

Find the two real roots of the equation.

since coefficients are real, $x = 1 + i$ is also a root; these give the factor $x^2 - 2x + 2$ (M1) A1

divide the quartic by $x^2 - 2x + 2$: quotient is $x^2 - 3$ (M1) A1

solve $x^2 - 3 = 0$ (M1)

$$x = \pm\sqrt{3} \approx \pm 1.73 \quad \text{A1}$$

[7 marks]